Chapter 3 - Rings and Moduls of Fractions

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Exercise 1

(i) ($\Rightarrow$): Let M be generated by $\{m_1, m_2, \dots, m_n\}$, for $\forall i \in [1, n] \cap \mathbb{Z}$, we have

$$\forall r, t \in S, \exists s \in S, \text{s.t. } m_i s(r-t) = 0, \text{i.e. } m_i s = 0$$

Denote the s above as s_i. Since M is finitely-generated, we can let

$$s = s_1 s_2 \dots s_n$$

And have $sM = 0$.

(ii) ($\Leftarrow$): When such s exist, then for $\forall \frac{m_1}{s_1}, \frac{m_2}{s_2} \in S^{-1}M$,

$$s(s_1 m_2 - s_1 m_2) = 0 \implies \frac{m_1}{s_1} = \frac{m_2}{s_1}$$

So $S^{-1}M = 0$.

$\square$

Exercise 2

By *Proposition 1.9*, the problem is equivalent to prove that for any $\frac{a}{s} \in S^{-1}A$, we have

$$\forall \frac{b}{t} \in S^{-1}A, 1 - \frac{a}{s}\frac{b}{t} \text{is a unit in } S^{-1}A$$

Since $1 + \mathfrak{a}$ is a multiplicative set, $\exists c \in \mathfrak{a}, \text{s.t. } st = 1 + c$. Now

$$\begin{aligned} 1 - \frac{a}{s}\frac{b}{t} &= 1 - \frac{ab}{st} \\ &= 1 - \frac{ab}{1+c} \\ &= \frac{1+c-ab}{1+c} \end{aligned}$$

And we can easily see $\frac{1+c}{1+c-ab} = 1 + \frac{ab}{1+c-ab} \in S^{-1}A$. So the inverse of $1 - \frac{a}{s}\frac{b}{t}$ is $1 + \frac{ab}{1+c-ab}$. $\square$

Exercise 3

Define

$$f : (ST)^{-1}A \longrightarrow U^{-1}(S^{-1}A)$$

by mapping

$$\frac{a}{st} \longmapsto \frac{\frac{a}{s}}{\frac{t}{1}}$$

We'll prove f is an isomorphism:

(i) Surjection: Trivial.

(ii) Injection:

$$\frac{\frac{a_1}{s_1}}{\frac{t_1}{1}} = \frac{\frac{a_2}{s_2}}{\frac{t_2}{1}}$$

$\Leftrightarrow$

$$\exists t \in T, \text{s.t. } \frac{t}{1}\left(\frac{a_1 t_2}{s_1} - \frac{a_2 t_1}{s_2}\right) = 0$$

$\Leftrightarrow$

$$\exists t \in T, \text{s.t. } t(a_1 t_2 s_2 - a_2 t_1 s_1) = 0$$

$\Leftrightarrow$

$$\frac{a_1}{s_1 t_1} = \frac{a_2}{s_2 t_2}$$

So f is injective.

(iii) Homomorphism: Trivial.

$\square$

Exercise 4

Define

$$f : S^{-1}B \longrightarrow T^{-1}B$$

by

$$\frac{b}{s} \longmapsto \frac{b}{f(s)}$$

We'll prove it's an isomorphism:

(i) Homomorphism: Trivial

(ii) Surjection: Trivial

(iii) Injection:

$$\frac{b_1}{f(s_1)} = \frac{b_2}{f(s_2)}$$

$\Leftrightarrow$

$$\exists s \in S, \text{s.t. } f(s)\left(b_1 f(s_2) - b_2 f(s_1)\right) = 0$$

$\Leftrightarrow$

$$\frac{b_1}{s_1} = \frac{b_2}{s_2}$$

$\square$

Exercise 5

(i) By *Proposition 3.11-(v)*, the operation S^{-1} commutes with the formations of radicals. So $\forall \mathfrak{p} \subseteq A$,

$$(\mathrm{Nil}(A))_{\mathfrak{p}} = \mathrm{Nil}(A_{\mathfrak{p}}) = 0$$

Therefore by *Proposition 3.8*, $\mathrm{Nil}(A_{\mathfrak{p}}) = 0$.

(ii) No. Counterexample given here

Exercise 6

(i) Σ is nonempty, therefore has maximal elements by *Zorn's Lemma*.

(ii) (I) When $S \in \Sigma$ is maximal, we'll prove $A \backslash S$ is a minimal prime ideal. $\forall x, y \in A \backslash S_0$, let

$$S_x = \left\{sx^n \mid s \in S, n \in \mathbb{Z}^+ \cup \{0\}\right\}, S_y = \left\{sy^n \mid s \in S, n \in \mathbb{Z}^+ \cup \{0\}\right\}$$

Since $S \in \Sigma$ is maximal and $S_x, S_y \supsetneq S$, we have $S_x, S_y \notin \Sigma, 0 \in S_x, 0 \in S_y$ Therefore $\exists s_1, s_2 \in S, n_1, n_2 \in \mathbb{Z}^+$, such that

$$s_1 x^{n_1} = s_2 x^{n_2} = 0$$

And we can see

$$s_1 s_2 (x+y)^{n_1+n_2-1} = 0, s_1 s_2 (xy)^{n_1 n_2} = 0$$

Which is to say $x + y, xy \in A \backslash S$, and we can easily obtian $A \backslash S$ is an ideal. The prime property is trivial, and the maximal property can be obtained directly from *(II)* and the maximal property of S.

(II) When $A \backslash S$ is a minimal prime ideal of A: Denote $I = A \backslash S$. For $\forall x, y \in S$, we have $x, y \notin I$. If $xy \in I$, by the prime property of I we will immediately obtian $x \in I$ or $y \in I$, which is a contradictory. So $xy \notin I, xy \in S$, S is a multiplicatively closed subset.

Also, if S is not maximal in Σ, the one in Σ strictly contains S will correspond to a smaller prime ideal in $A \backslash S$, which contradicts with the minimal property of I.

$\square$

Exercise 7

(i) (I) $(\Leftarrow)$: When $A \backslash S$ is a union of prime ideals, $x \in A \backslash S$ or $y \in A \backslash S$ will immediately implies $xy \notin S$. So $xy \in S$ will imply $x, y \in S$.

(II) $(\Rightarrow)$: We'll prove that any elements in $A \backslash S$ was contained in a prime ideal which is disjoint to S: $\forall a \in A \backslash S, (a) \cap S \subseteq \{0\}$, otherwise if $a^n = a \cdot a^{n-1} \in S$, we'll have $a \in S$, contradictory.

Now we consider Σ be the set of all ideals containing (a) and being disjoint to S. By *Zorn's Lemma*, we can pick out a maximal element, namely $\mathfrak{p} \subseteq \Sigma$. We'll prove $\mathfrak{p}$ is prime:

Suppose we have $xy \in \mathfrak{p}, x, y \notin \mathfrak{p}$. Since $\mathfrak{p}$ is maximal and $\mathfrak{p} + (x), \mathfrak{p} + (y) \supsetneq \mathfrak{p}$, we know both of $\mathfrak{p} + (x), \mathfrak{p} + (y)$ intersects with S. Therefore $\exists s_1, s_1 \in S$ such that $\exists p_1, p_2 \in \mathfrak{p}, a_1, a_2 \in A$,

$$s_1 = p_1 + a_1 x, s_2 = p_2 + a_2 x$$

Then we have

$$(s_1 - p_1) s_2 \in S$$

whereas

$$\begin{aligned}(s_1 - p_1) s_2 &= a_1 x (p_2 + a_2 y) \\ &= a_1 p_2 x + a_1 a_2 xy \in \mathfrak{p}\end{aligned}$$

Which means $(s_1 - p_1) s_2 \in S \cap \mathfrak{p}$, contradicting with the construction of $\mathfrak{p}$.

$\square$

(ii) By *(i)*, $\overline{S}$ defined by the complement in A of union of prime ideals disjoint from S is indeed a saturated multiplicative closed subset of A.

We only have to prove $\overline{S}$ is the unique minimal one, which is trivial since the family of sets, which consists of prime ideals disjoint from S, has a unique maximal element (final object, where morphisms taken under inclusion).

(iii)

$$\mathfrak{p} \cap S \subsetneq \{0\}$$

$\Leftrightarrow$

$$\exists p \in \mathfrak{p}, a \in \mathfrak{a}, \text{s.t. } 1 + p = a$$

$\Leftrightarrow$

$$\mathfrak{a} + \mathfrak{p} = (1)$$

$\Leftrightarrow$

$$\mathfrak{a}, \mathfrak{p} \text{ are coprime}$$

Conversely, it's easy to prove that $\mathfrak{a}, \mathfrak{p}$ being coprime will implies $\mathfrak{p} \cap S \subsetneq \{0\}$. Therefore $\overline{S}$ is the complement of the union of all prime ideals which are not coprime with $\mathfrak{a}$.

Exercise 8

(I) *(i)* $\Rightarrow$ *(ii)*: $\forall t \in T, \phi(\frac{t}{1}) = \frac{t}{1}$ is a unit in $T^{-1}A$. Since ϕ is bijective, the preimage of the inverse of $\frac{t}{1} \in T^{-1}A$ is also the inverse of $\frac{t}{1} \in S^{-1}A$.

(II) *(ii)* $\Rightarrow$ *(iii)*: $\forall t \in T$, $\frac{t}{1}$ is a unit in $S^{-1}A$ and we denote $\frac{a}{s} = \left(\frac{t}{1}\right)^{-1}$. Therefore $\exists s_0 \in S, \text{s.t. } s_0(at-s) = 0$, which indicate

$$(s_0 a)\, t = s_0 s \in S$$

Thus we can let $x = s_0 a$.

(III) *(iii)* $\Rightarrow$ *(iv)*: Suppose $\exists t \in T, \text{s.t. } t$ is contained in a prime ideal $\mathfrak{p}$ which is disjoint from S. By *(iii)*, $\exists x \in A, \text{s.t. } xt \in S$. However, $t \in \mathfrak{p} \Rightarrow xt \in \mathfrak{p} \Rightarrow xt \in S \cap \mathfrak{p}$, contradictory.

(IV) *(iv)* $\Rightarrow$ *(v)*: Otherwise T cannot be contianed in $\overline{S}$.

(V) *(v)* $\Rightarrow$ *(ii)*: Suppose $\frac{t}{1}$ is not a unit in $S^{-1}A$. Then there exist a maximal ideal $\subseteq S^{-1}A$, s.t. $\frac{t}{1} \in \mathfrak{m}$.

Consider the natural homomorphism $f : A \longrightarrow S^{-1}A$ and denote $f^{-1}(\mathfrak{m}) = \mathfrak{p}$. It immediately follows that $\mathfrak{p} \cap S = \emptyset$ (otherwise $f(\mathfrak{p} = S^{-1}A)$). Also, from $\frac{t}{1} \in \mathfrak{m}$ we have $t \in \mathfrak{p}$, i.e. $\mathfrak{p}$ meets T, therefore $\mathfrak{p}$ meets S, contradictory.

(VI) *(ii)* $\Rightarrow$ *(i)*: Elements in $T^{-1}A$ generally has the form $\frac{a}{t}$. Since $\frac{t}{1}$ is a unit in $S^{-1}A$, we have

$$a \cdot \left(\frac{t}{1}\right)^{-1} \in S^{-1}A, \phi\left(a \cdot \left(\frac{t}{1}\right)^{-1}\right) = \frac{a}{t}$$

hence ϕ is surjective.

On the other hand, suppose $\phi\left(\frac{a}{s}\right) = 0$ where $\frac{a}{s} \neq 0$. Then $\exists t \in T \backslash S, at = 0$. In this case $\frac{t}{1}$ cannot be unital in $S^{-1}A$.

$\square$

Exercise 9

(I) We follow from the hint: Denote the family of all multiplicatively closed set which doesn't contains 0 by Σ, the problem is equivalent to prove that all the maximal elements in Σ must contains S.

We prove by contradictory and suppose there exist a maximal element $S \in \Sigma$ such that $\exists a \in S_0 \backslash S$. Let $S_a = \{sa^n \mid s \in S, n \in \mathbb{Z}^+ \cup \{0\}\}$, which is a multiplicatively closed set strictly contains S. Because S is maximal in Σ, $S_a \notin \Sigma, 0 \in S_a$. Hence, $\exists n \in \mathbb{Z}^+, \text{s.t.}$

$$sa^n = 0$$

However, from $a \in S_0 \backslash S$ we know a^n is a non-zero-divisor, i.e. $s = 0$, which is absurd. Hence the proof was done. $\square$

(II)
- (i) Homomorphism $A \longrightarrow S_0^{-1}A$ is injective iff $\forall s \in S_0, a \in A \backslash \{0\}, sa \neq 0$. This is to say S_0 consists of non-zero-divisors.
- (ii) Every element in $A \backslash S_0$ will be mapped to a zero-divisor while every element in S_0 will be mapped to a unit, obviously.
- (iii) Injectivity is obtained from *(i)*, and the preimage of $\frac{a}{s} \in S^{-1}A$ is $as^{-1} \in A$, obviously.

$\square$

Remark 1

From **Exercise 3.6, 3.9**, we can see saturated sets and sets made up of prime-ideals-union acts as complements to each other. And if we let S be maximal, the corresponding $A\backslash S$ will be left to be only one prime idea.

Exercise 10

(ii) **Hinited**

(i) By *Proposition 2.19(iv)*, for any $S^{-1}A$-module N, if we have $S^{-1}A$-module M, M' and homomorphism $f : M \longrightarrow M'$ such that
$$0 \to M \xrightarrow{f} M'$$
exact as $S^{-1}A$-module sequence, we only need to prove
$$0 \longrightarrow S^{-1}M \otimes_{S^{-1}A} S^{-1}N \xrightarrow{f\otimes 1} S^{-1}M' \otimes_{S^{-1}A} S^{-1}N$$
is exact.

Notice that restriction on the rings under which the modules are only affect the structure of scalar multiplication while holding the image and kernel, we may view N, M, M' as A-module and f as A-module homomorphism while having
$$0 \to M' \xrightarrow{f|_A} M$$
exact as A-module sequence.

Therefore from the absolute flatness of A we may have N as a flat A-module, i.e.
$$0 \longrightarrow M' \otimes_A N \xrightarrow{f|_A\otimes 1} M \otimes_A N$$
is exact as a A-module sequence.

Now by *Proposition 3.3*, the $S^{-1}\_$ functor is exact, and
$$0 \longrightarrow S^{-1}\left(M' \otimes_A N\right) \xrightarrow{S^{-1}(f|_A\otimes 1)} S^{-1}\left(M \otimes_A N\right)$$
is an exact $S^{-1}A$ sequence. Moreover, by *Proposition 3.7* such sequence is isomorphic to
$$0 \longrightarrow S^{-1}M \otimes_{S^{-1}A} S^{-1}N \xrightarrow{f\otimes 1} S^{-1}M \otimes_{S^{-1}A} S^{-1}N$$
and we get its exactness.

(ii) (a) When A is absolutely flat: From *(i)* $A_{\mathfrak{m}}$ is flat for any maximal ideal $\mathfrak{m} \subseteq A$. And by *Exercise 2.28*, an absolutely flat local ring must be a field, i.e. $A_{\mathfrak{m}}$ is a field.

(b) When for any maximal ideal $\mathfrak{m} \subseteq A$, $A_{\mathfrak{m}}$ is a field: For any A-module N, consider $N_{\mathfrak{m}}$ as a $A_{\mathfrak{m}}$-module. Since $A_{\mathfrak{m}}$ is a field, $N_{\mathfrak{m}}$ is a vector space, therefore has a basis and is a free module over $A_{\mathfrak{m}}$. It's well-known that a free module must be flat, therefore $N_{\mathfrak{m}}$ is flat over $A_{\mathfrak{m}}$. Because $\mathfrak{m}$ is arbitarily chosen, by *Proposition 3.10* we know N is flat, and therefore A is absolutely flat.

□

Exercise 11

(I) (a) $(i) \Leftrightarrow (ii)$:

$(i) \Leftrightarrow$
$$\left(A/\operatorname{Nil} A\right)_{\mathfrak{m}} \text{ is a field for all maximal ideals } \mathfrak{m} \text{ of } A/\operatorname{Nil} A$$
$\Leftrightarrow$
$$\left(A/\operatorname{Nil} A\right)_{\mathfrak{m}} \text{ only has ideals}(1), (0) \text{ for all maximal ideals } \mathfrak{m} \text{ of } A/\operatorname{Nil} A$$
$\Leftrightarrow$
$$\left(A/\operatorname{Nil} A\right)_{\mathfrak{m}} \text{ only has prime ideal } (0) \text{ for all maximal ideals } \mathfrak{m} \text{ of } A/\operatorname{Nil} A$$

$\Leftrightarrow$

$$A/\operatorname{Nil} A \text{ has an unique prime ideal contianed in } \mathfrak{m} \text{ for all maximal ideals } \mathfrak{m} \text{ of } A/\operatorname{Nil} A$$

$\Leftrightarrow$

$$\text{Every prime ideals of } A/\operatorname{Nil} A \text{ is maximal}$$

$\Leftrightarrow (ii)$

(b) $(ii) \Rightarrow (iii)$: When all prime ideals are maximal ideals, then every subsets, which consists of single point, are the final objects in the category of $\operatorname{Spec} A$ (morphisms under inclusion), therefore closed.

(c) $(iii) \Rightarrow (ii)$: From definition, T_1 property implies every single point set is closed. And by the definition of *Zarisky Topology*, every point is maximal

(d) $(i) \Rightarrow (iv)$: Notice: the cannonical map $A \longrightarrow A/\operatorname{Nil} A$ induced a homeomorphism $\operatorname{Spec} B \longrightarrow \operatorname{Spec} A$. And to prove $\operatorname{Spec} A$ Hausdorff, we only need to prove $\operatorname{Spec} B$ Hausdorff.

First we'll find an idempotent element, which is the key to constrcut distinct open sets: Let $B = A/\operatorname{Nil} A$. From the assumption of (i) we know B is absolutely flat. And we've proved $(i) \Leftrightarrow (ii)$ so for any $\mathfrak{p}, \mathfrak{q} \in \operatorname{Spec} B, \exists b \in \mathfrak{p} \backslash \mathfrak{q}$. By **Lemma 1**, $\exists x \in B,$ s.t. $b = b^2 x$. Now let $e = bx$ and we have

$$e^2 = b^2 x^2 = bx = e$$

and e is idempotent. Moreover, since $\mathfrak{p}$ is prime, we have $x \in \mathfrak{p} \Rightarrow e \in \mathfrak{p} \Rightarrow (b) = (e)$.

The $\mathfrak{p}, \mathfrak{q} \in \operatorname{Spec} B$ was arbitarily chosen, and we can always have an idempotent $e \in \mathfrak{p}$. Now consider the open sets

$$\operatorname{Spec} B \backslash V(e), \operatorname{Spec} B \backslash V(1-e)$$

Becuase $e \notin \mathfrak{q}, \mathfrak{q} \in \operatorname{Spec} B \backslash V(e)$. Also, $1 - e \notin \mathfrak{p}$ (otherwise both $e, 1-e \in \mathfrak{p}$ and $1 = e + 1 - e \in \mathfrak{p}, \mathfrak{p} = (1)$ which is a contradictory) so $\mathfrak{p} \in \operatorname{Spec} B \backslash V(1-e)$. It's only left for us to prove

$$(\operatorname{Spec} B \backslash V(e)) \cap (\operatorname{Spec}(B) \backslash V(1-e)) = \emptyset$$

which is trivial since

$$\begin{aligned}
(\operatorname{Spec} B \backslash V(e)) \cap (\operatorname{Spec}(B) \backslash V(1-e)) &= \operatorname{Spec} B \backslash \left(V(e) \cup V(1-e)\right) \\
&= \operatorname{Spec} B \backslash V(e(1-e)) \\
&= \operatorname{Spec} B \backslash V(e - e^2) \\
&= \operatorname{Spec} B \backslash V(0) \\
&= \emptyset
\end{aligned}$$

(e) $(iv) \Rightarrow (iii)$: Immediately obtained from the seperation theorem in topology ($T_2 \Rightarrow T_1$).

(II) (i) *Compactness*: We'll use B to prove since they're homeomorphic: For a family of groups of elements in B $\{E_i\}_{i \in \mathcal{I}}$, suppose we have such open cover

$$\operatorname{Spec}(B) = \bigcup_{i \in \mathcal{I}} \operatorname{Spec} B \backslash V(E_i)$$

we'll prove there exist a finite subcover. Notice

$$\begin{aligned}
\bigcup_{i \in \mathcal{I}} \operatorname{Spec} B \backslash V(E_i) &= \operatorname{Spec} B \backslash \bigcap_{i \in \mathcal{I}} V(E_i) \\
&= \operatorname{Spec}(B) \backslash V\left(\bigcup_{i \in \mathcal{I}} E_i\right) \text{ (by \textit{Exercise 1.15(iii)})}
\end{aligned}$$

Therefore $\operatorname{Spec}(B) = \bigcup_{i \in \mathcal{I}} \operatorname{Spec} B \backslash V(E_i) \Leftrightarrow V\left(\bigcup_{i \in \mathcal{I}} E_i\right) = \emptyset$. And to obtain a finite subcover, we only have to find out a finite subset $E_0 \subseteq E,$ s.t. $V(E_0) = \emptyset$. It's very easy to achieve, since one can just pick out two element in $\mathfrak{p} \backslash \mathfrak{q}, \mathfrak{q} \backslash \mathfrak{p}$ where $\mathfrak{p}, \mathfrak{q} \in \operatorname{Spec}(B), \mathfrak{p} \neq \mathfrak{q}$.

(ii) *Totally disconnected*: We follow from the proff of $(i) \Rightarrow (iv)$: We've proved $\operatorname{Spec} B \backslash V(e)$ and $\operatorname{Spec} B \backslash V(1-e)$ are disjoint. Also, from

$$\begin{aligned}(\operatorname{Spec} B \backslash V(e)) \cup (\operatorname{Spec} B \backslash V(1-e)) &= \operatorname{Spec} B \backslash \left(V(e) \cap V(1-e)\right) \\ &= \operatorname{Spec} B \backslash V(\{e, 1-e\}) \\ &= \operatorname{Spec} B \backslash V(1) \\ &= \operatorname{Spec} B\end{aligned}$$

we know $\operatorname{Spec} B \backslash V(e)$ and $\operatorname{Spec} B \backslash V(1-e)$ is a seperation of $\operatorname{Spec} B$. Now for any subspace of $\operatorname{Spec} B$ with more than 2 elements, we can endow it with the corresponding subset topology and get a seperation of the subspace, i.e. any subspace with more than 2 elements are disconnected.

□

Lemma 1

For any absolutely flat ring A and $\forall a \in A, \exists x \in A, \text{s.t. } a = a^2 x$.

Proof. Consider any ideal $I, J \subseteq A$. Since A is absolutely flat, A/I, as an A-module, is flat. Now we consider the exact sequence

$$0 \to J \xrightarrow{i} A$$

where i is the inclusion map. We have

$$0 \longrightarrow A/I \otimes_A J \xrightarrow{i \otimes 1} A/I \otimes_A A$$

exact, i.e.

$$0 \longrightarrow J/IJ \xrightarrow{i \otimes 1} j + I$$

exact. Consider the kernel of $i \otimes 1$ (which is IJ be exactness): Since

$$i \otimes 1 : j + IJ \longmapsto j + I$$

we know $\operatorname{Ker} i \otimes 1 = I \cap J$. Hence $IJ = I \cap J$, and this works for any ideals in A Now let $I = J = (a)$. We have $(a) = (a^2)$, So $\exists e \in A, \text{s.t. } a = a^2 e$. □

Exercise 12

It's trivial that torsion elements of M form a submodule of M.

(i) Suppose $\exists x \notin T(M), \text{s.t. } \exists a \in A, a\overline{x} = 0, \text{i.e. } ax \in T(M)$. Then $\exists b \in A, \text{s.t. } b(ax) = 0$, therefore $abx = 0, x \in T(M)$, contradictory.

(ii) $\forall m \in T(M), \exists a \in A, \text{s.t. } am = 0 \Rightarrow af(m) = 0, f(m) \in T(N)$.

(iii) *(My proof might be a bit ambiguous, but I think I did proved it out.)*

Denote

$$0 \xrightarrow{f} M' \xrightarrow{g} M \xrightarrow{h} X''$$

and we need to prove

$$0 \xrightarrow{\overline{f}} T(M') \xrightarrow{\overline{g}} T(M) \xrightarrow{\overline{h}} T(X'')$$

is exact.

First, $\operatorname{Ker} \overline{h} = \operatorname{Ker} h \cap T(M'') = \operatorname{Im} g \cap T(M'')$. To prove $\operatorname{Ker} \overline{h} = \operatorname{Im} \overline{g}$, we need to show

$$g(T(M')) = g(M') \cap T(M)$$

i.e. $g(M' \backslash T(M')) \cap T(M) \subseteq \{0\}$.

Suppose $\exists m' \in M' \backslash T(M'), g(m') \in T(M)$. Then $\exists a \in A, ag(m') = 0 \Rightarrow g(am') = 0 \Rightarrow am' \in \operatorname{Ker} g$ Which is to say

$$am' \in \operatorname{Ker} g = \operatorname{Im} f = \{0\}$$

Contradictory. Hence we have $\operatorname{Ker} \overline{h} = \operatorname{Im} \overline{g}$.

Also, the case $\operatorname{Ker} \overline{g} = \operatorname{Im} \overline{f}$ is trivial.

(iv) *(The hint seems to be complicated, I'll just follow my own thought.)*

Denote

$$\varphi : M \longrightarrow K \otimes_A M$$
$$x \longmapsto 1 \otimes x$$

First, for all $m \in T(M)$, suppose $am = 0$. Then

$$\varphi(m) = 1 \otimes m = \frac{1}{a} \otimes am = 0$$

So $T(M) \subseteq \operatorname{Ker} \varphi$.

On the other hand, $\forall m \in \operatorname{Ker} \varphi, \varphi(m) = 1 \otimes m = 0$ A is an integral domain, so there're no zero divisors in K. Thus $1 \otimes m$ must be annihilated in the m term, i.e. $\exists a \in A, am = 0 \Rightarrow m \in T(M)$.

□

Exercise 13

(I) On one hand, $\forall \frac{m}{s} \in S^{-1}M, \exists a \in A,$ s.t. $a \cdot \frac{m}{s} = 0 \Rightarrow \frac{am}{s} = 0, m \in T(M), \frac{m}{s} \in S^{-1}(T(M))$. So $S^{-1}M \subseteq S^{-1}(T(M))$.

On the other hand, for all $\frac{m}{s} \in S^{-1}(T(M))$: By $m \in T(M)$ we have $a \in A,$ s.t. $ma = 0$. So $a \cdot \frac{m}{s} = 0, \frac{m}{s} \in S^{-1}M$.

(II) By *(I)*,

$$T(M_\mathfrak{p}) = (A \backslash \mathfrak{p})^{-1} T(M) = (T(M))_\mathfrak{p}$$
$$T(M_\mathfrak{m}) = (A \backslash \mathfrak{m})^{-1} T(M) = (T(M))_\mathfrak{m}$$

Use *Proposition 3.8* and we're done.

□

Exercise 14

We follow the hint: By *Exercise 2.2*, $(A/\mathfrak{a}) \otimes_A M \cong M/\mathfrak{a}M$. So we have $M_{\overline{\mathfrak{m}}} = 0$ for all maximal ideals $\overline{m}$ in $A/\mathfrak{a}$, which correspond to $\mathfrak{m}$ in A. Therefore by *Proposition 3.8*, we have $M/\mathfrak{a}M = 0,$ i.e. $M = \mathfrak{a}M$. □

Exercise 15

(Following the hint given, the idea is that we want to use the theories in vector spaces to solve this problem, so we would try to turn the original modules into vector spaces.)

We use the method in the hint directly: Without loss of generality let A be local and $\mathfrak{m} \subseteq A$ is the unique maximal ideal. Let ϕ be the surjective homomorphism in the hint and define $N = \operatorname{Ker} \phi$. Now we have exact sequence

$$0 \longrightarrow N \longrightarrow F \xrightarrow{\phi} F \longrightarrow 0$$

Since $\operatorname{Tor}_1^A(k, F) = 0$, we have

$$0 \longrightarrow k \otimes_A N \longrightarrow k \otimes_A F \xrightarrow{1 \otimes \phi} k \otimes_A F \longrightarrow 0$$

exact.

$k \otimes_A F \cong k^n$ is a n-dimensional vector space, $1 \otimes \phi$ is surjective, hence bijective, i.e. $k \otimes_A N = 0$. Therefore $N/\mathfrak{m}N \cong (A/\mathfrak{m}) \otimes_A N = k \otimes_A N = 0$, so by *Nakayama's Lemma*, $N = 0$, ϕ is an isomorphism. □

Exercise 16

(a) $(i) \Rightarrow (ii)$: Denote the homomorphism $A \longrightarrow B$ by φ. $\forall \mathfrak{p} \in \operatorname{Spec} A, \mathfrak{p}^{ec} = \mathfrak{p}$. So by *Proposition 3.16*, $\mathfrak{p}$ is the contraction of a prime ideal of B.

(b) $(ii) \Rightarrow (iii)$: For any maximal ideal $\mathfrak{m}$ there exist a prime ideal of B, namely $\mathfrak{p}$, such that $\mathfrak{m} = \mathfrak{p}^c$. So $\mathfrak{m}^e = B\varphi(\mathfrak{m}) = \mathfrak{p} \subsetneq (1)$.

(c) ()

Lemma 2

If the morphism $A \longrightarrow B$ is faithfully flat, then any finitely generated A-module M, M' with sequence

$$0 \longrightarrow M' \longrightarrow M$$

is exact iff

$$0 \longrightarrow M' \otimes_A B \longrightarrow M \otimes_A B$$

is exact.

Proof. For any A-module homomorphism $f : M' \longrightarrow M$, we have

$$0 \longrightarrow \operatorname{Ker} f \longrightarrow M' \xrightarrow{f} M$$

exact. By tensor product we have

$$0 \longrightarrow \operatorname{Ker} f \otimes_A B \longrightarrow M' \otimes_A B \xrightarrow{f \otimes 1} M \otimes_A B$$

exact.

(i) When $0 \longrightarrow \operatorname{Ker} f \longrightarrow M' \xrightarrow{f} M$ is exact:
Now $\operatorname{Ker} f = 0$, therefore

$$\operatorname{Ker} f \otimes 1 = \operatorname{Ker} f \otimes_A B = 0.$$

Done.

(ii) When $0 \longrightarrow \operatorname{Ker} f \otimes_A B \longrightarrow M' \otimes_A B \xrightarrow{f \otimes 1} M \otimes_A B$ is exact:
Now $\operatorname{Ker} f \otimes_A B = 0$. Denote the natural homomorphism $\phi : A \longrightarrow B$, and we have $B \cong A/\operatorname{Ker}\phi$,

$$\begin{aligned} \operatorname{Ker} f \otimes_A B &\cong \operatorname{Ker} f \otimes_A (A/\operatorname{Ker}\phi) \\ &= \operatorname{Ker} f / \operatorname{Ker}\phi \operatorname{Ker} f \end{aligned}$$

i.e. $\operatorname{Ker} f / \operatorname{Ker} f \operatorname{Ker}\phi = 0$. By *Proposition 3.8* we may suppose A is local, at which $\operatorname{Ker}\phi \subseteq \operatorname{Jab} A$. Also, from M, M' being finitely generated we know $\operatorname{Ker} f$ is finitely generated. Hence, by *Nakayama's Lemma* $\operatorname{Ker} f = 0$, i.e.

$$0 \longrightarrow M' \xrightarrow{f} M$$

exact.

□

Exercise 17

Since $g \circ f$ is flat, for any finitely generated A-module M, M' and exact sequence

$$0 \longrightarrow M' \longrightarrow M$$

we have

$$0 \longrightarrow M' \otimes_A C \longrightarrow M \otimes_A C$$

exact. From *Exercise 2.15* we know

$$M \otimes_A C = M \otimes_A (B \otimes_B C) \cong (M \otimes_A B) \otimes_B C$$

and so does M', therefore

$$0 \longrightarrow (M' \otimes_A B) \otimes_B C \longrightarrow (M \otimes_A B) \otimes_B C$$

is exact. Now from **Lemma 2** we have

$$0 \longrightarrow M' \otimes_A B \longrightarrow M \otimes_A B$$

exact. Now use *Proposition 2.19* and we're done. □

Exercise 18

From *Proposition 3.10*, $B_{\mathfrak{p}}$ is flat over $A_{\mathfrak{p}}$.

Also, consider the image of $B\backslash\mathfrak{q}$ after localization $B \longrightarrow B_{\mathfrak{p}}$, denoted by $\overline{B\backslash\mathfrak{q}}$. Then the result of localization

$$B \longrightarrow B_{\mathfrak{p}} = (A\backslash\mathfrak{p})^{-1}B \longrightarrow (\overline{B\backslash\mathfrak{q}})^{-1}B_{\mathfrak{p}}$$

is isomorphic to the localization of B under the multiplicative subset generated by $f(A\backslash\mathfrak{p})$ and $B\backslash\mathfrak{q}$, which is $B\backslash\mathfrak{q}$ since $f(A\backslash\mathfrak{p}) \subseteq B\backslash\mathfrak{q}$. Therefore $B_{\mathfrak{q}}$ is flat over $B_{\mathfrak{p}}$.

Combining $B_{\mathfrak{p}}$ flat over $A_{\mathfrak{p}}$ and $B_{\mathfrak{q}}$ flat over $B_{\mathfrak{p}}$, we get $B_{\mathfrak{q}}$ flat over $A_{\mathfrak{p}}$. Becuase $\mathfrak{p} = \mathfrak{q}^c$, use *Exercise 16(i),(ii)* and we're done. □

Exercise 19

(i) (a) When $M \neq 0$: Use *Proposition 3.8*.

(b) When $\operatorname{Supp} M \neq \emptyset$: Suppose $M = 0$, then $\operatorname{Supp} M = \emptyset$, contradictory.

(ii)

$$(A/\mathfrak{a})_{\mathfrak{p}} \neq 0$$

$\Leftrightarrow$

$$(A\backslash\mathfrak{p})^{-1}(A/\mathfrak{a}) \neq 0$$

$\Leftrightarrow$

$$\forall x \in A\backslash\mathfrak{p}, xA \nsubseteq \mathfrak{a}$$

$\Leftrightarrow$

$$\forall x \in A\backslash\mathfrak{p}, x \notin \mathfrak{a}$$

$\Leftrightarrow$

$$(A\backslash\mathfrak{p}) \cap \mathfrak{a} = \emptyset$$

$\Leftrightarrow$

$$\mathfrak{a} \subseteq \mathfrak{p}$$

$\Leftrightarrow$

$$\mathfrak{p} \subseteq V(\mathfrak{a})$$

□

(iii) Denote

$$0 \to M' \to M \xrightarrow{\varphi} M'' \to 0.$$

We have

$$M' \cong \operatorname{Ker}\varphi, M'' = \operatorname{Im}\varphi$$

By *Corollary 3.6*, for any $\mathfrak{p} \in \operatorname{Spec} A$ we have

$$0 \longrightarrow M'_{\mathfrak{p}} \longrightarrow M_{\mathfrak{p}} \xrightarrow{\varphi\otimes 1} M''_{\mathfrak{p}} \longrightarrow 0$$

exact. And the proof becomes trivial after noticing those facts.

(iv) Trivial.

(v) Trivial.

(vi) Trivial.

(vii)

$$\begin{aligned}\operatorname{Supp} M/\mathfrak{a}M &= \operatorname{Supp}\left((A/\mathfrak{a}) \otimes_A M\right)\\ &= \operatorname{Supp}(M) \cap \operatorname{Supp}(A/\mathfrak{a})\\ &= V(\operatorname{Ann} M) \cap V(\operatorname{Ann} A/\mathfrak{a})\\ &= V(\mathfrak{a} + \operatorname{Ann} M)\end{aligned}$$

(viii) **(SKIPPED TEMPORARILY)**

(a) $\forall \mathfrak{q} \in \operatorname{Supp}(B \otimes_A M)$, we have $(B \otimes_A M)_{\mathfrak{q}} \neq 0$, i.e.

$$(B \backslash \mathfrak{q})^{-1} (B \otimes_A M) \neq 0$$

Therefore

$$\left(A \backslash f^{-1}(\mathfrak{q})\right) M \neq 0$$

Which is to say $\mathfrak{q} \in f^{*-1}(\operatorname{Supp} M)$. Hence $\operatorname{Supp}(B \otimes_A M) \subseteq f^{*-1}(\operatorname{Supp} M)$.

(b) $\forall \mathfrak{p} \in \operatorname{Supp} M$, we need to prove $(B \otimes_A M)_{\mathfrak{p}} \neq 0$, i.e.

$$B_{\mathfrak{q}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} \neq 0$$